Assignment-06

Data Set

Machine	Temperature	Vibration	Failure
1	High	High	Yes
2	High	Low	Yes
3	Medium	High	Yes
4	Low	Low	No
5	Low	Medium	No
6	Medium	Low	No
7	High	Medium	Yes
8	Low	High	No

Calculate Total Entropy

Variable : Failure

Y = 4

No = 4

Total = 8

$$\text{entropy}(S) = -\sum P_i \log_2 P_i$$

$$= -P_1 \log_2 P_1 - P_2 \log_2 P_2$$

$P(Yes) = 4/8 = 0.5$

$P(No) = 4/8 = 0.5$

$$\text{Entropy}(S) = -(0.5)\log_2(0.5) - (0.5)\log_2(0.5)$$

$$= -(0.5)(-1) - (0.5)(-1)$$

Date
Page

Entropy(S) = 0.5 + 0.5
= 1

Calculate Information gain for Temperature

Temperature value:

A) High

Yes = 3
No = 0
Entropy (High) = 0

B) Medium

Yes = 1
No = 1
Entropy (Medium) = $-(0.5)\log_2(0.5) - (0.5)\log_2(0.5)$
= 1

C) Low

Yes = 0
No = 3
Entropy (Low) = 0

Weighted Entropy for Temperature

W.Entropy = $\frac{3}{8}(0) + \frac{2}{8}(1) + \frac{3}{8}(0)$

= 0.25

Information Gain for temperature

Gain = Entropy (s) - W.Entropy

= 1 - 0.25

= 0.75

Information Gain for Vibration.

A) High

Yes = 2

NO = 1

P (Yes) = 2/3

P (No) = 1/3

Entropy (High) = $-\frac{2}{3} \log_2 2/3 - 1/3 \log_2 1/3$

= 0.918

B) Low

Yes = 1

NO = 2

Entropy (Low) = 0.918

c) Medium

Yes = 1

No = 1

entropy (Medium) = 1

Weighted entropy for vibration

W.Entropy = $3/8 (0.918) + 3/8 (0.918) + 2/8 (1)$

$= 0.344 + 0.344 + 0.25$

$= 0.938$

Information gain

Gain = $1 - 0.938$

$= 0.062$

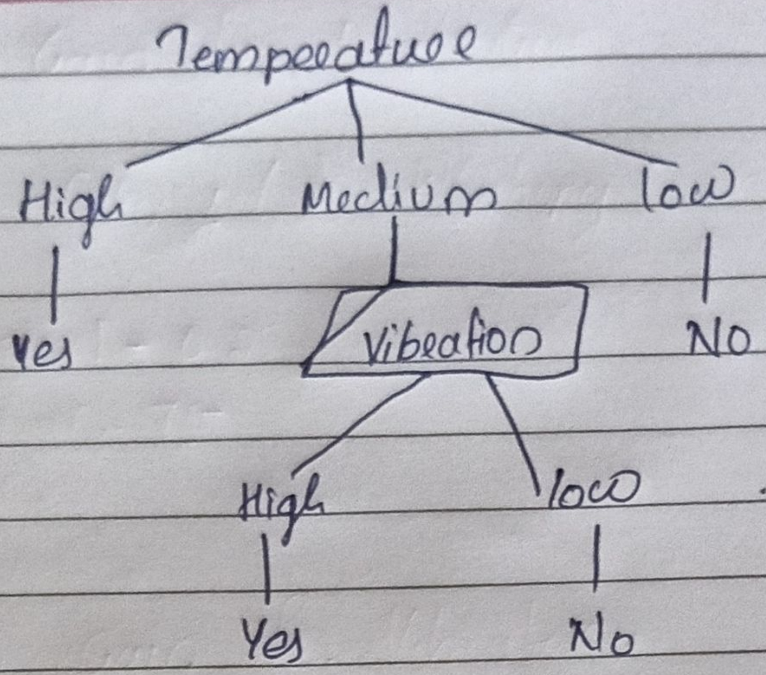
Choose root node

Temperature	0.75
Vibration	0.062

∴ root node : Temperature

Decision Tree

Temperature = High → Yes
Temperature = low → No
Temperature = Medium → Vibration = High → Yes
Vibration = Low → No.



Prediction

for the new machine:

Temperature = High
Vibration = Medium

$\therefore$ Yes

Machine will fail = Yes.